

Spinal pathways, touch and pain

The written notes for the spinal pathways, touch and pain walkthrough. Week 5 competencies 5, 13 and 14.

Inside the spinal cord

- Gray matter, in the center, shaped like a butterfly: cell bodies, dendrites and synapses.
- Dorsal horn: receives sensory input. Ventral horn: cell bodies of somatic motor neurons. Lateral horn, in the thoracic and upper lumbar cord: autonomic neurons.
- White matter, around the gray: myelinated axons in tracts, grouped into dorsal, lateral and ventral columns.
- Dorsal root: sensory axons in. Its swelling, the dorsal root ganglion, holds the sensory cell bodies. Ventral root: motor axons out.
- The two roots join into one spinal nerve, which carries both. There are 31 pairs.
- Damage to a dorsal root ganglion (shingles) affects sensation in one strip of skin. Damage to the ventral horn (polio) causes weakness with sensation spared.

Three long pathways

Tract	Carries	Direction	Where it crosses the midline	First synapse
Dorsal column	Fine touch, vibration, proprioception	Up	Medulla	Medulla (dorsal column nuclei)
Spinothalamic	Pain, temperature	Up	Spinal cord, within a segment or two of entry	Dorsal horn of the cord
Corticospinal	Voluntary movement	Down	Bottom of the medulla (the pyramids)	Ventral horn motor neuron

- Both sensory pathways relay in the thalamus before reaching the somatosensory cortex.
- Corticospinal neurons in the cortex are upper motor neurons; the motor neurons in the ventral horn are lower motor neurons.

- Each side of the brain senses and moves the opposite side of the body.

Brown-Séquard syndrome: a cut through one half of the cord

- Same side as the cut, below it: loss of fine touch, vibration and position sense, and loss of voluntary movement.
- Opposite side, below it: loss of pain and temperature.
- The split pattern locates the lesion in the cord. Above the medulla all three pathways have crossed, so a brain lesion takes everything from the same side of the body.

Touch, from skin to cortex

- Touch ends in the primary somatosensory cortex, the postcentral gyrus, laid out as a map of the body.
- The somatosensory homunculus is disproportionate: cortical area matches receptor density and how finely a part discriminates, not its size. Lips and fingertips get the most.

Pain and its modulation

	Fast pain	Slow pain
Fiber	A-delta: thin, myelinated	C: thinnest, unmyelinated
Arrives	First	About a second later
Feels	Sharp, prickling, easy to locate	Dull, aching or burning, hard to locate, lasting

Referred pain

Pain from an organ felt in the skin. Visceral and somatic afferents converge on the same spinal projection neurons, and the brain assigns the signal to the skin. Heart pain felt in the left arm is the example.

Gate control theory

Large touch fibers excite inhibitory interneurons in the dorsal horn that inhibit the pain projection neuron. Rubbing an injury reduces the pain signal.

Endogenous opioids

Enkephalins, endorphins and dynorphins. Released by interneurons activated by descending pathways from the

brainstem. They act before the synapse, reducing calcium entry and transmitter release (substance P, glutamate) from

the nociceptor, and after it, opening potassium channels to hyperpolarize the projection neuron.